

# Introduction to Loops

## Understanding Iteration using Flowcharts

Programming Fundamentals Team

SETU

2026-09-14

# Outline

1. Why do we need loops?
2. Selection versus Iteration
3. Anatomy of a loop
4. Infinite loops
5. Useful loop patterns
6. Activities

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# 1.1 What if we needed to repeat this?

[ Without loops, repeated tasks quickly become messy:

```
print("Welcome")  
print("Welcome")  
print("Welcome")  
print("Welcome")
```

A loop lets us write the repeated instruction once. ],

# 1.1 What if we needed to repeat this?

## Key idea

Computers are very good at repetition.

## 1.2 Everyday repetition

- Checking each student's mark
- Taking orders until the queue is empty
- Asking for a password until it is correct
- Displaying a menu until the user chooses Exit
- Reading temperatures from a sensor

**Loops are used when a set of instructions must be repeated.**

## 1.3 What is iteration?

**Iteration means repeating a set of instructions until a condition is met.**

Examples:

- Continue asking until the input is valid.
- Continue counting until the target is reached.
- Continue processing until there is no more data.

# Outline

1. Why do we need loops?

**2. Selection versus Iteration**

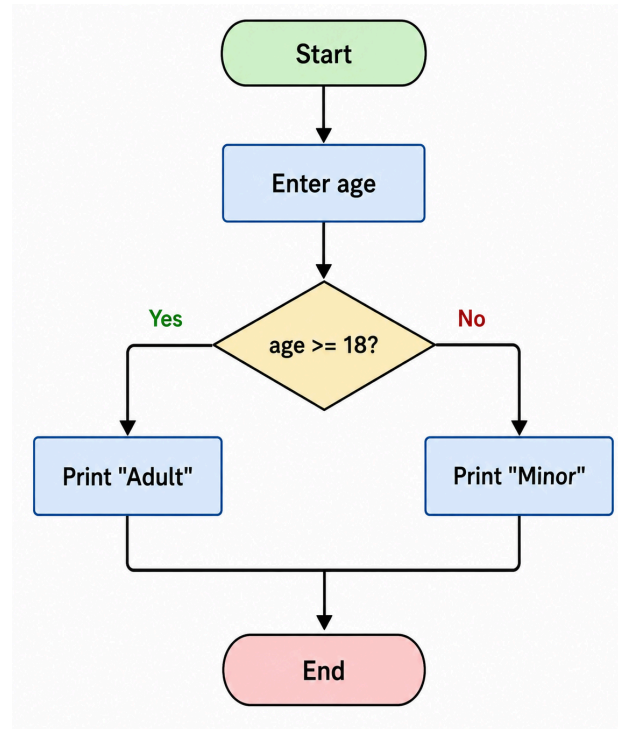
3. Anatomy of a loop

4. Infinite loops

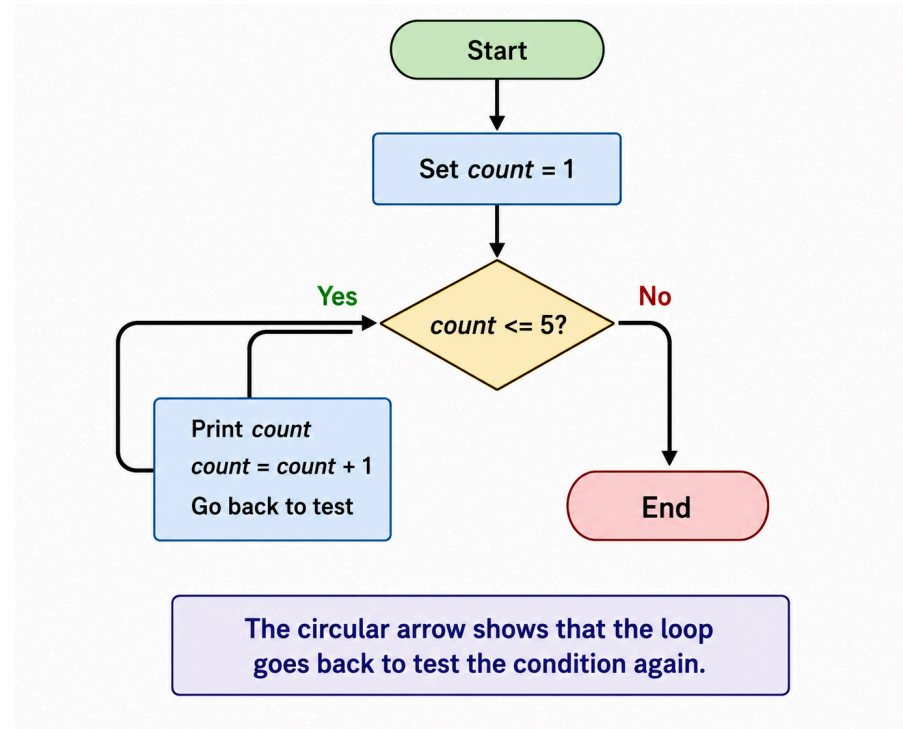
5. Useful loop patterns

6. Activities

## 2.1 Selection asks once



## 2.2 Iteration asks repeatedly



## 2.3 The key difference

### Selection

- Condition checked once
- One path is chosen
- Program continues

## 2.3 The key difference

### Iteration

- Condition checked repeatedly
- Instructions may run many times
- Stops when condition becomes false

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## 3.1 Every loop needs four things

### 1. Starting value



Example: `count = 1`

### 2. Condition



Example: `count <= 5`

### 3. Repeated action



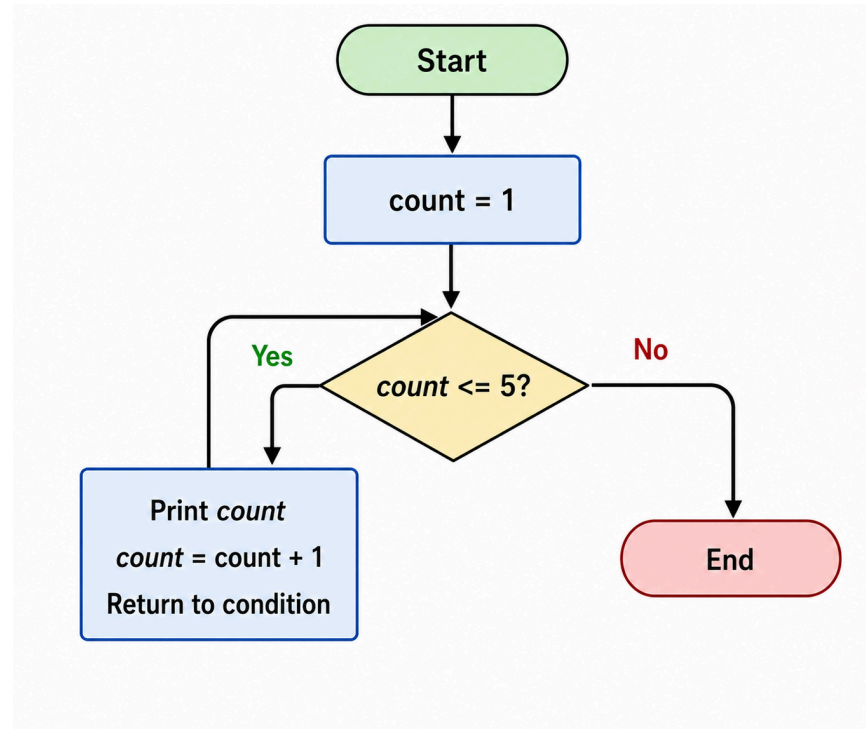
Example:  
print the number

### 4. Update / change



Example:  
`count = count + 1`

## 3.2 Counting to five: flowchart



## 3.3 Trace the loop

| count | Condition       | Output |
|-------|-----------------|--------|
| 1     | 1 <= 5 is True  | 1      |
| 2     | 2 <= 5 is True  | 2      |
| 3     | 3 <= 5 is True  | 3      |
| 4     | 4 <= 5 is True  | 4      |
| 5     | 5 <= 5 is True  | 5      |
| 6     | 6 <= 5 is False | Stop   |

## 3.4 Why does the loop stop?

The loop stops because the value changes.

count eventually becomes 6.

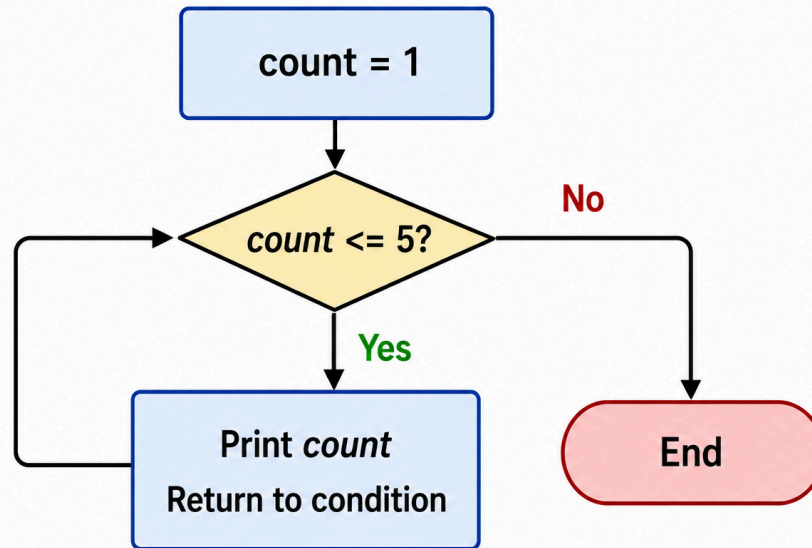
The condition `count <= 5` becomes false.

**Important:** a loop normally needs something inside it that moves it closer to stopping.

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## 4.1 What if we forget the update?



**count never changes, so the loop never ends.**

## 4.2 Infinite loop in code

```
count = 1

while count <= 5:
    print(count)
```

Problem: `count += 1` is missing.

## 4.3 Corrected code

```
count = 1

while count <= 5:
    print(count)
    count += 1
```

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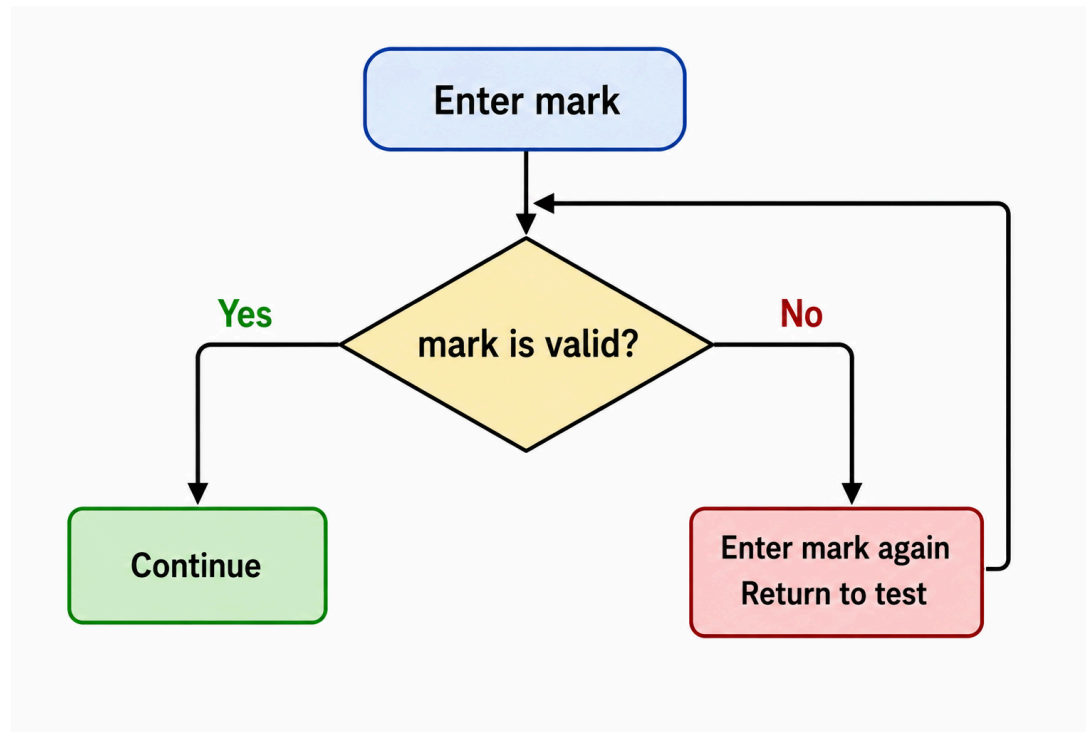
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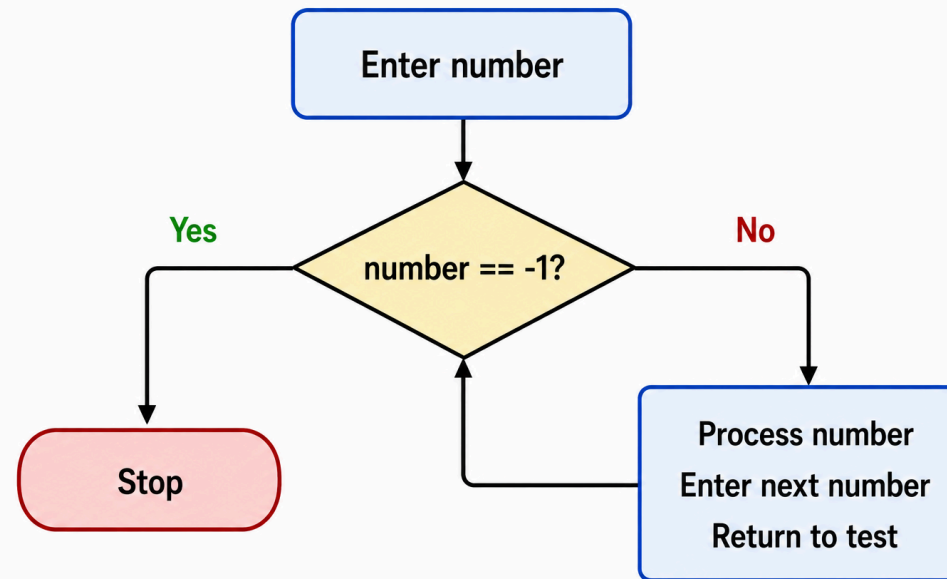
## 5.1 Validation loop

Use a loop when the user must enter valid data.



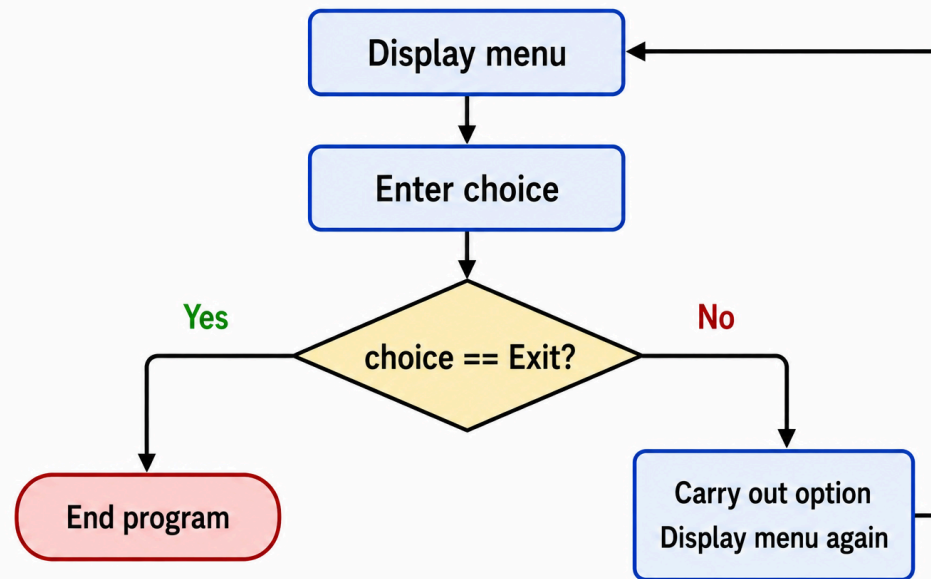
## 5.2 Sentinel loop

A **sentinel** is a special value that means “stop”.



## 5.3 Menu loop

Many programs repeat a menu until the user chooses **Exit**.



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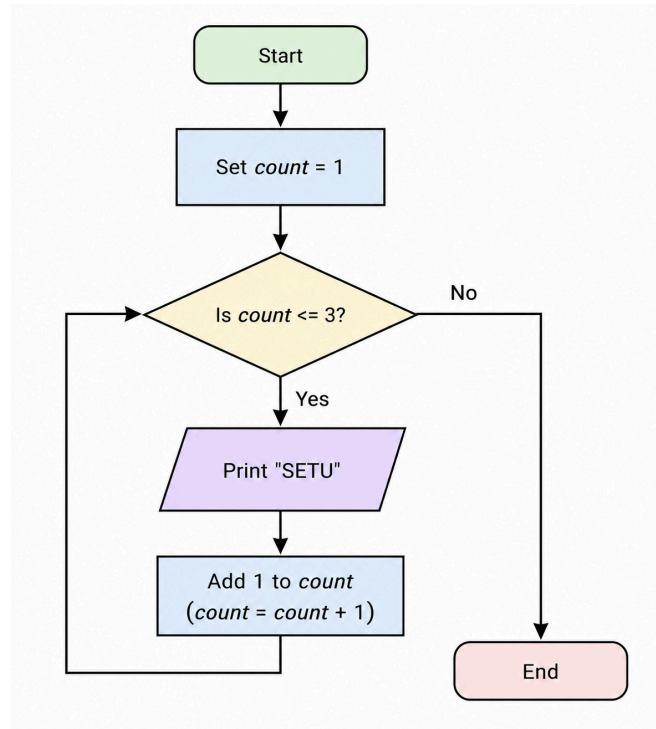
## 6.1 Activity: find the four parts

For the flowchart, identify:

1. Starting value
2. Condition
3. Repeated action
4. Update or stopping action

**This prepares you to write the Python version in the next lecture.**

## 6.2 Flowchart



## 6.3 Predict the output

Flowchart summary:

- Set count = 1
- While count <= 3
- Print "SETU"
- Add 1 to count

**Question:** what is printed?

## 6.4 Summary

- Iteration means repetition.
- Flowcharts help us plan loops before writing code.
- Selection asks once; iteration asks repeatedly.
- Every loop needs a condition and a way to stop.
- Missing updates can cause infinite loops.

**Next:** implementing these ideas in Python using `while` loops.

# Thanks for Watching - Any questions?

